

recheck your code for mistakes (errors apart from syntax errors – i.e. typos) that may have crept in while you were programming.

Here are some of the programming blunders that can cause errors:

### 1. **Incorrect syntax**

This is one syntax error that cannot be detected by the ‘Verify’ step.

Let’s say you want to use the comparison operator `==` (i.e. two equal signs) to compare value like this:

- `if (i == 4)`

and instead you write:

- `if (i = 4)`

Using the `=` (single equal sign) assigns a value of 4 to variable `i` and the statement is evaluated to ‘true’ causing huge unnoticeable errors.

### 2. **Overflow errors**

Overflow errors are errors caused when the limit of variables are exceeded. E.g. Let’s say you write a code with variable `i` declared as an integer.

The range of integer for Arduino™ UNO board with 16-bit microcontroller is -32,768 to 32,767.

So if you exceed 32767 or go below -32,768, overflow occurs and the microcontroller can’t count further and starts using ‘false’ value.

These are some invisible blunders that are neither easily noticed nor are they highlighted by the Verify tool.

## **To conclude...**

In this chapter, we saw how to install Arduino™ software, make the Arduino™ environment ready to upload sketches, and verify and upload sketches. We then looked at a few programs to get the LED to blink, and for use with the display, sensors and various shields. We then looked at the syntax used in the Arduino™ environment that’s an add-on to the general C language to achieve hardware control. And if your sketch doesn’t respond as you intended, you can refer to the final part of this chapter that points you to some possible errors that may have sneaked in.

We wish you the best of luck in putting the pieces of code in the right place. 